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Thermal management and performance augmentation of photovoltaic module subjected to actual dynamic conditions with passive cooling techniques

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Abstract

The paper analyses the combined effect of fin number (n) and inclination (β) on the overall performance of the PV module in the finned PV-PCM system by numerical simulation. System geometries with no fin to four fins are considered with inclinations varying from 0° to 45° to obtain different configurations of the PV-PCM system. Performance parameters such as mean temperature, PV efficiency, PV power output, mean velocity, and temperature contours are simulated for all the system configurations. The system is subjected to actual dynamic conditions such as variable solar radiation and ambient temperature. The maximum temperature reduction is obtained for the three-finned system with an inclination of 45°. The inclusion of fins is relatively effective for horizontal systems compared to other inclinations, and the highest relative enhancement in efficiency of 5.56% is obtained for horizontal PV-PCM system with 4 fins. Three-finned PV-PCM system with a 45° inclination angle produces the highest instantaneous PV electrical power output of 165.87 W and the lowest peak PV temperature of 50.23 °C, which is the optimum. The peak average velocity of melted PCM is maximum for unfinned PV-PCM system with $\beta = 45°$, indicating the presence of the highest strength of convection current in melted PCM.

Keywords PV-PCM system · Numerical simulation · Inclination · Fins · Power output

List of symbols

A	Porosity operator
A_s	Surface area of the PV panel (m^2)
B	Liquid fraction
c_p	Specific heat capacity (J kg^{-1} K^{-1})
C_m	Porosity constant
D	Dirac delta function
F	View factor
G	Incident solar radiation (W m^{-2})
g	Acceleration due to gravity (m s^{-2})
Gr	Grashof number
h	Heat transfer coefficient (W m^{-2} K^{-1})
k	Thermal conductivity (W m^{-1} K^{-1})
L_c	Characteristic length (m)
L_h	Latent heat (J kg^{-1})
m	Number of days
n	Number of fins
P	Pressure (Pa)
P_{out}	Power output of PV (W)
Pr	Prandtl Number
$\dot{Q}_g$	Rate of heat generation (W m^{-3})
Ra	Rayleigh number
Re	Reynolds number
S	Secondary volumetric forces (N m^{-3})
T	Temperature (°C)
T_f	Film temperature (°C)
T_m	Transition temperature (°C)
T_{sky}	Sky temperature (°C)
t	Time (s)
u	Velocity in the x-direction (m s^{-1})
v	Velocity in the y-direction (m s^{-1})
V	Volume (m^3)
v_w	Wind velocity (m s^{-1})

Greek symbols

α	Thermal expansion coefficient (K^{-1})
β	Inclination angle of PV with horizontal (°)
γ_c	Solar irradiance coefficient
γ_w	Wind azimuth angle (°)
δ	Declination angle (°)

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η	Efficiency of PV(%)
ε	Emissivity of material
ν	Kinematic viscosity ($m^2\ s^{-1}$)
μ	Dynamic viscosity (Pa s)
ψ	Latitude (°)
ρ	Density (kg m^{-3})
σ	Stefan–Boltzmann constant (W $m^{-2}\ K^{-4}$)
θ	Elevation angle (°)
$(\tau\alpha)$	Absorptivity–transmissivity product for glass cover and PV cell

Subscripts

a	Air
al	Aluminium
amb	Ambient
avg	Average
b	Bottom
c	Critical
free	Free convection
forced	Forced convection
gl	Glass
gr	Ground
l	Liquid
pcm	PCM
s	Solid
si	Silicon
t	Top
te	Tedlar

Abbreviations

BIPV	Building integrated photovoltaic
EVA	Ethylene vinyl acetate
MEPCM	Micro-encapsulated phase change material
PCM	Phase change material
PV	Photovoltaic

Introduction

The entire world is now contending with the adverse impact of climate change, primarily caused by developmental activities and industrial expansion. Therefore, the role of green energy in growing electricity demand and rapidly increasing overall energy demand becomes more crucial [1]. The prospects of the future world's energy scenario will rest on renewable sources. All over the world, renewable energy markets have been multiplying since the last decade. Compared to all other renewable energy resources, solar energy has the highest potential. Solar energy can be converted to thermal energy using devices like collectors, or it can produce electricity directly using photovoltaic principles with solar cells. The optimum temperature for the solar PV module is 25 °C, as it can absorb the sun's radiation with maximum efficiency [2]. Absolute conversion efficiency and output of the crystalline silicon cell decreased by 0.08% and 0.45%, respectively, for every 1 °C increase in temperature from the optimum [3]. Apart from this, a sharp temperature rise can progressively damage the cell encapsulation's lamination, which leads to corrosion. It is necessary to employ proper cooling methods to reduce the effects of high temperatures. Dubey et al. [4] presented the correlations between the electric performance of Si-based solar cells with temperature. It was concluded that no single correlation works for all the applications, and the choice of the equation depends on factors like mounting geometry or building integration level. Kaldellis et al. [5] examined the effect of wind speed on the performance parameters of the PV, such as operating temperature, thermal loss factor, frequency, and efficiency. The results showed that the difference between the cell and ambient temperatures decreased with increased wind speed, which validates the importance of cooling.

Many researchers have recently started focussing on passive cooling techniques based on PCMs, which store the energy as latent heat during the phase transition with very small or no temperature rise [6, 7]. A standard solid–liquid PCM with a melting range close to room temperature is bonded to the rear side of the PV module, resulting in a PV-PCM system that leads to decreased PV surface temperature and improved performance [8, 9]. There are many techniques to improve heat transfer within PCMs because they have limited thermal conductivity [10]. Attaching fins to the PCM container is one of them. Fins boost the rate of heat transfer and manage the temperature of the PV module [11, 12]. At the same time, an increase in surface area with fins may lead to increased resistance to the flow of liquid PCM and a reduction in heat transfer by convection. Hence, there is a limit to which fins can be added to PCM.

The effect of paraffin wax as a PCM on the performance of concentrated PV was analysed by Ceylan et al. [13] using a mathematical model. While the temperature of the standalone CPV was found to be well above 100 °C, the paraffin-filled PV system's temperature was found to be between 80 and 100 °C. The effect of a microencapsulated PCM (MEPCM) saturated with water on the PV surface temperature was numerically examined by Ho et al. [14]. They found that when a 5 cm MEPCM layer with a melting temperature of 30 °C was combined with a conventional PV module, the PV efficiency was found to be increased by 3%, with a maximum temperature of 32.8 °C. Sharma et al. [15] conducted an experimental study for 2 h on the BIPV-PCM system using RT42 as PCM and compared the results with naturally ventilated BIPV without PCM. Temperature was found to be reduced by 12 °C at 1000 W m^{-2}. Improvements of 7.7%, and 4.4% were observed for electrical efficiency and open circuit voltage, respectively, with PCM usage. Thermal and economic analysis of passive cooling of PV-PCM systems was conducted by Japs et al. [16]. The PCM having

higher thermal conductivity results in higher performance, and a temperature reduction of 10 °C was found for 4–6 h when a PCM RT35 was coupled with PV. Mahdi et al. [17] conducted a numerical investigation of different PCMs for achieving passive thermal management of PV modules at different tilt angles. The findings indicated that enhancing the tilt angle from 0° to 60° can enhance thermal management efficacy by a maximum of 15%.

Kant et al. [18] numerically simulated the PV performance with RT35HC as PCM. The impact of wind speed and tilt angle of the PV-PCM system were noted, and compared with the PV-only case, and a drop in the temperature by 3 °C was observed. Stropnik and Stritih [19] presented a method to improve the PV electrical efficiency and output by incorporating RT28HC as PCM. Experimental results showed an improvement in the temperature of 35.6 °C on the surface of PV with PCM compared with bare PV. Khanna et al. [20] numerically analysed the influence of system inclination, wind velocity, wind direction, and outdoor temperature on the performance of PV-PCM. Results showed that as the inclination increased from 0° to 90°, the reduction in temperature was 10 °C, and efficiency was enhanced by 1%. As per the investigation carried out by Pichandi et al. [21], the incorporation of eutectic inorganic PCM into the PV module resulted in an average power output improvement of 17.63 W. They also reported a surface temperature decrease of up to 10 °C in the modules, contributing to improved performance.

A 0.06 m thick bed of metal-embedded paraffin wax was used as PCM by Maiti et al. [22] for the purpose of thermal regulation of PV. The PV temperature was maintained at 60–65 °C for 3 h, leading to an overall gain of 55%. Huang et al. [23] used RT27 as the PCM to investigate the effects of convection current on the heat transfer performance of a finned PV-PCM system. Compared to PV-PCM without fins, finned PV-PCM was found to keep the temperature below 35 °C efficiently. Tanuwijava et al. [24] performed a CFD analysis to study the effect of the macro aspect ratio of MEPCM for PV applications during transient fluctuations of daily solar irradiation using two PCMs with melting temperatures of 26 °C and 34 °C. The system consisting of RT25 PCM linked to the PV panel was numerically studied by Biwole et al. [25]. The panel temperature was maintained at less than 40 °C for 80 min.

Abdulmunem and Jalil [26] conducted a study based on using paraffin wax and aluminium fins to regulate the temperature of PV cells and enhance their performance. The results indicated that the performance of the PV cell was enhanced when coupled with PCM/Fins, resulting in a decrease in temperature of approximately 14.19%. According to the numerical analysis by Waqas and Ji [27], the temperature of conventional PV modules without PCM can reach up to 64 °C, but using PCM-filled shutters with a transition point of 35 °C can lower the temperature to 42 °C. Hasan et al. [28] compared PV-PCM with salt hydrate (PCM1) and a eutectic fatty acid mixture (PCM2) with internal fins in two distinct climates. It was concluded that, in hot and stable climate conditions such as Vehari, PCM1 produced a higher drop in PV temperature of 3–4 °C, resulting in 3% more power savings. Singh et al. [29] used numerical methods to examine the effectiveness of using PCM heat sinks and finned PCM heat sinks for cooling PV systems. Their results indicated that finned PCM heat sinks could increase PV cooling by 19 K. Al-Rabghi [30] developed a mathematical model to investigate the behaviour of PV modules with a finned heat sink on their back under different operational conditions. The simulation results indicated that the optimal fin spacing and height were 12 mm and 150 mm, respectively. The implementation of these fins resulted in an expected reduction in cell temperature of around 25% and a corresponding increase in efficiency of about 10.8%.

The key takeaway from the literature review is that while temperature control of PV panels using PCM is an appealing option compared to other active PV cooling approaches, PCM's limited thermal conductivity is a crucial drawback. To improve the heat transfer rate in the system, appropriate methods, such as the inclusion of fins, must be used. On the one hand, the presence of fins enhances the PCM's thermal conductivity, increasing heat transfer rates and thereby more effectively regulating PV module temperature. On the other hand, it adversely affects the convection heat transfer inside melted PCM, especially for inclined PV-PCM systems, due to increased flow resistance. The study of these two opposing factors when fins are introduced in the PV-PCM system and how they affect the overall system performance in realistic working conditions has not been attempted so far.

In the current work, an effort has been made to investigate the combined influence of fin number and inclination on the performance of the finned PV-PCM system under actual weather conditions, i.e. variable solar radiation and ambient temperature. Two-dimensional mathematical modelling and transient numerical simulation of different PV-PCM configurations are carried out in the present work to analyse the combined effect of inclination ($\beta = 0°$, 15°, 30° and 45°) and fin number ($n = 0$, 1, 2, 3, and 4) on the performance of the PV-PCM system. The optimum fin number-inclination combination is found from the simulation results. Performance parameters of the PV module, such as average temperature, electrical efficiency, electric power output, and characteristics of PCM, such as temperature contours and magnitude of the average velocity of melted PCM, are simulated for

all the configurations of the PV-PCM system considered. The performance of the finned PCM systems is compared with the conventional PV-PCM system (unfinned) with the parameters simulated.

Physical system

The schematic of the PV module coupled with finned PCM considered in the present work is represented in Fig. 1. The PV module considered for the current work is established with the polycrystalline PV, composed of five thin layers of materials: Glass, EVA, silicon, EVA and Tedlar, whose properties are mentioned in Table 1. The PCM RT35HC is attached to the backside of the PV by using two aluminium sheets of a thickness of 2 mm. Fins of suitable numbers can be placed between the two plates along the length of the system to achieve various system configurations. Fins are extended to the full depth of the PCM in all the arrangements. The inclination angle is represented by β, which is the angle made by the PV-PCM system with the horizontal. The system can be operated at different inclinations for a particular number of fins, and with this, different operational configurations are available.

Table 1 Properties of PV layers [33]

Material	ρ/kg m^{-3}	c_p/J kg^{-1} K^{-1}	k/W m^{-1} K^{-1}	Thickness/m
Glass	3000	500	1.8	0.003
EVA	960	2090	0.35	0.0005
Silicon	2320	678	148	0.0003
Tedlar	1210	1250	0.2	0.0001
Al	2675	903	211	0.002

Mathematical modelling

The mathematical modelling of the two-dimensional finned PV-PCM domain for all configurations is carried out by considering certain assumptions. These are listed as follows:

1. the distribution of solar radiation is uniform on the front surface of the PV for all configurations at any instant.
2. the PCM flow is incompressible, two-dimensional, unsteady, laminar and Newtonian.
3. contact thermal resistances between different layers of the system are considered to be negligible.
4. PV components are considered isotropic and homogenous.
5. PV resistive losses are negligible.

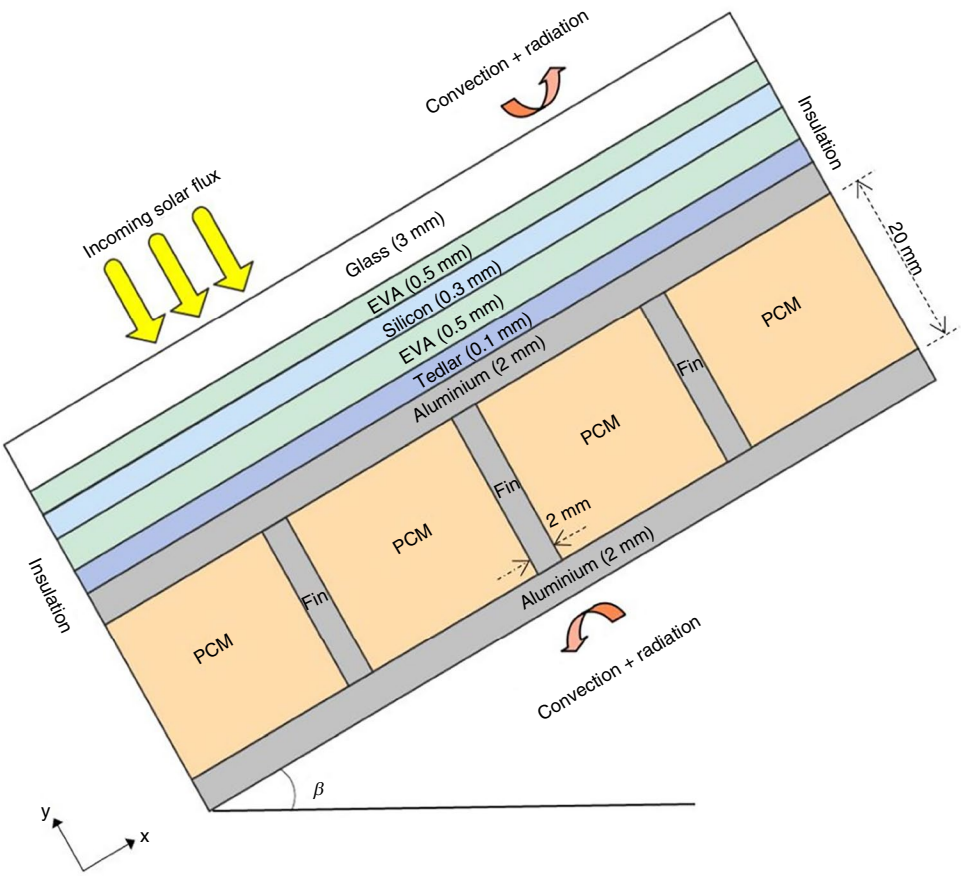


Fig. 1 Schematic of the finned PV-PCM system with dimensions

Governing equations

Governing differential equations for two-dimensional temperature distribution (in the x–y plane) of all the layers of the PV module is

$$\rho c_{\mathrm{p}} \frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) + \dot{Q}_{\mathrm{g}} \tag{1}$$

where $\dot{Q}_{\mathrm{g}}$ is the rate of heat generation per unit volume in the module in W m^{-3} which is expressed with [18]

$$\dot{Q}_{\mathrm{g}} = \left[1 - \eta_{\mathrm{PV}}\right](\tau\alpha) G_{\mathrm{module}} \frac{A_{\mathrm{s}}}{V} \tag{2}$$

G_{module} is the solar irradiation in W m^{-2} and $(\tau\alpha)$ is the transmissivity–absorptivity product of the glass layer.

Different relations that connect G_{module}, $G_{\mathrm{horizontal}}$ and G_{incident} are [18]

$$G_{\mathrm{horizontal}} = G_{\mathrm{incident}} \sin\theta \tag{3}$$

$$G_{\mathrm{module}} = \frac{G_{\mathrm{horizontal}} \sin(\theta + \beta)}{\sin\theta} \tag{4}$$

$$\theta = 90 - \psi + \delta \tag{5}$$

and

$$\delta = 23.45 \sin\left[\frac{360}{365}(284 + m)\right] \tag{6}$$

where θ, β, ψ, and δ are the elevation, inclination, latitude and declination angles of the PV. η_{PV} is the PV electrical efficiency, which varies with temperature [31]. m is the number of days.

$$\eta_{\mathrm{PV}} = 20\left[1 - 0.005\left(T_{\mathrm{PV}} - 25\,^{\circ}\mathrm{C}\right) + 0.085 \ln\left(G_{\mathrm{module}}(t)/1000\right)\right] \tag{7}$$

The governing differential equations for PCM, including natural thermo-gravitational melt, have the following forms:

Continuity:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{8}$$

x-momentum:

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial P}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + S_{\mathrm{x}} + \rho g_{\mathrm{x}} \alpha_{\mathrm{pcm}}\left(T - T_{\mathrm{m}}\right) \tag{9}$$

y-momentum:

$$\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = -\frac{\partial P}{\partial y} + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) + S_{\mathrm{y}} + \rho g_{\mathrm{y}} \alpha_{\mathrm{pcm}}\left(T - T_{\mathrm{m}}\right) \tag{10}$$

Thermal energy:

$$\rho c_{\mathrm{p}} \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x} - \rho c_{\mathrm{p}} u T\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y} - \rho c_{\mathrm{p}} v T\right) \tag{11}$$

The additional source terms S_{x} and S_{y} which are secondary volumetric forces added to help the solver quickly to achieve PCM velocities to zero value in the solid phase, and they can be defined in laminar flow conditions as [25]

$$S_{\mathrm{x}} = -A(T) \cdot \vec{u};\ \ S_{\mathrm{y}} = -A(T) \cdot \vec{v} \tag{12}$$

where $A(T)$ is a porosity operator which is defined as [32].

$$A(T) = \frac{C_{\mathrm{m}}(1 - B(T))^2}{B(T)^3 + \varepsilon} \tag{13}$$

When the liquid fraction is zero, the secondary volumetric force dominates all other source terms in the momentum equations, and this results in speedy convergence of the solver to PCM velocities equal to zero. In liquid PCM, $A(T)$ is equal to zero. C_{m} is a constant that indicates a mushy medium, and its value depends on the morphology of the PCM. For organic PCM, the value of C_{m} is generally taken as 10^5 [25]. ε is a very small integer (=0.0001) to avoid the denominator to become 0. $B(T)$ is the liquid fraction or melt fraction, which can also be used to define the variation of thermal properties during phase transition and is defined as [33]

$$B(T) = \begin{cases} 0 & \text{if } T < \left(T_{\mathrm{m}} - \Delta T\right) \\ \left(T - T_m + \Delta T\right)/(2\Delta T) & \text{if } \left(T_{\mathrm{m}} - \Delta T\right) \le T \le \left(T_{\mathrm{m}} + \Delta T\right) \\ 1 & \text{if } T > \left(T_{\mathrm{m}} + \Delta T\right) \end{cases} \tag{14}$$

The density and thermal conductivity of PCM are modelled as [34, 35]

$$\rho_{\mathrm{pcm}}(T) = \rho_{\mathrm{solid}} + \left(\rho_{\mathrm{liquid}} - \rho_{\mathrm{solid}}\right) B(T) \tag{15}$$

$$k_{\mathrm{pcm}}(T) = k_{\mathrm{solid}} + \left(k_{\mathrm{liquid}} - k_{\mathrm{solid}}\right) B(T) \tag{16}$$

The modified heat capacity of the PCM introduces the added energy requirement through the phase transition. The function $D(T)$ is involved in this, which is a Gaussian function or Dirac delta function and can be defined as [25, 35]

$$D(T) = e^{\frac{-(T - T_{\mathrm{m}})^2}{(\Delta T/4)^2}} \Big/ \sqrt{\pi(\Delta T/4)^2} \tag{17}$$

Except for the interval $(T_{\mathrm{m}} - \Delta T)$ and $(T_{\mathrm{m}} + \Delta T)$, the value of the function $D(T)$ is zero. The integral of $D(T)$

over the transition region is one in order to fulfil the energy balance in the transition region. The final modified heat capacity is the mixture of sensible heat and latent heat of PCM, which can be expressed as

$$c_{\mathrm{p\,pcm}}(T) = c_{\mathrm{p\,solid}} + \left(c_{\mathrm{p\,liquid}} - c_{\mathrm{p\,solid}}\right)B(T) + L_{\mathrm{h}}D(T) \quad (18)$$

The viscosity of the PCM is modelled with the help of $A(T)$, which gives the right value when the PCM is in the liquid phase and forces the PCM to behave as a solid by providing an extremely large viscosity as [25]

$$\mu(T) = \mu_{\mathrm{l}}(1 + A(T)) \quad (19)$$

where μ_{l} is the dynamic viscosity of liquid PCM.

For the two-dimensional temperature distribution of aluminium plates and fins, the governing differential equation is

$$\rho c_{\mathrm{p}}\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) \quad (20)$$

Boundary and initial conditions

For the front surface of the PV, the boundary condition is

$$\begin{aligned} k_{\mathrm{gl}}\left(\frac{\partial T}{\partial y}\right) &= \left(h_{\mathrm{t,free}} + h_{\mathrm{t,forced}}\right)\left(T_{\mathrm{amb}} - T_{\mathrm{gl}}\right) \\ &+ \sigma\varepsilon_{\mathrm{gl}}F_{\mathrm{t\text{-}sky}}\left(T_{\mathrm{sky}}^4 - T_{\mathrm{gl}}^4\right) \\ &+ \sigma\varepsilon_{\mathrm{gl}}F_{\mathrm{t\text{-}ground}}\left(T_{\mathrm{gr}}^4 - T_{\mathrm{gl}}^4\right) + \alpha_{\mathrm{gl}}G_{\mathrm{module}} \end{aligned} \quad (21)$$

where $h_{\mathrm{t,free}}$ and $h_{\mathrm{t,forced}}$ are natural and forced heat transfer coefficients, respectively, and are defined as [31]

$$h_{\mathrm{t,free}} = \begin{cases} \left[0.13\left\{(\mathrm{Gr\,Pr})^{1/3} - (\mathrm{Gr_c\,Pr})^{1/3}\right\} + 0.56(\mathrm{Gr_c\,Pr}\sin\beta)^{1/4}\right]k_{\mathrm{a}}/L_{\mathrm{c}} & \text{if } \beta > 30° \\ \left[0.13(\mathrm{Gr\,Pr})^{1/3}\right]k_{\mathrm{a}}/L_{\mathrm{c}} & \text{if } \beta \le 30° \end{cases} \quad (22)$$

The term $\mathrm{Gr_c}$ is the critical Grashof number, and it is evaluated with

$$\mathrm{Gr_c} = 1.327 \times 10^{10}\exp\left(-3.708\left(\frac{\pi}{2} - \beta\right)\right) \quad (23)$$

The top forced convection heat transfer coefficient is defined as [31]

$$h_{\mathrm{t,forced}} = 0.848k_{\mathrm{a}}\left[\sin\beta\cos\gamma v_{\mathrm{w}}\,\mathrm{Pr}/\nu_{\mathrm{a}}\right]^{0.5}\left(L_{\mathrm{c}}/2\right)^{-0.5} \quad (24)$$

where the angle $\gamma = \left|\gamma_{\mathrm{pv}} - \gamma_{\mathrm{a}}\right|$ is the wind azimuth angle which is the horizontal projection from the normal to the windward side of the system.

$\gamma_{\mathrm{pv}} = 0$, for the PV-PCM system facing south orientation and $\gamma = \gamma_{\mathrm{w}}$.

T_{amb}, T_{gl} and T_{sky} are ambient, glass and sky temperatures, respectively. Sky temperature is evaluated with [36],

$$T_{\mathrm{sky}} = 0.037536T_{\mathrm{amb}}^{1.5} + 0.32T_{\mathrm{amb}} \quad (25)$$

$F_{\mathrm{t\text{-}sky}}$ and $F_{\mathrm{t\text{-}gr}}$ are the shape factors between the top face of the PV module with the sky and the ground, respectively, and these are evaluated with

$$F_{\mathrm{t\text{-}sky}} = \frac{1 + \cos\beta}{2},\ F_{\mathrm{t\text{-}gr}} = \frac{1 - \cos\beta}{2} \quad (26)$$

The sidewalls of the PV-PCM system are thermally insulated, so

$$\left(\frac{\partial T}{\partial x}\right) = 0 \quad (27)$$

With respect to different layers of the PV module, the interface boundary condition is

$$k_{\mathrm{r}}\left(\frac{\partial T_{\mathrm{r}}}{\partial y}\right) = k_{\mathrm{s}}\left(\frac{\partial T_{\mathrm{s}}}{\partial y}\right) \quad (28)$$

where r and s are two separate layers of materials on either side of the interface.

At the bottom side of the PV-PCM system, considering radiation and convection losses [18],

$$\begin{aligned} k_{\mathrm{al}}\left(\frac{\partial T}{\partial y}\right) &= \left(h_{\mathrm{b,free}} + h_{\mathrm{b,forced}}\right)\left(T_{\mathrm{amb}} - T_{\mathrm{al}}\right) \\ &+ \sigma\varepsilon_{\mathrm{al}}F_{\mathrm{b\text{-}sky}}\left(T_{\mathrm{sky}}^4 - T_{\mathrm{al}}^4\right) \\ &+ \sigma\varepsilon_{\mathrm{al}}F_{\mathrm{b\text{-}gr}}\left(T_{\mathrm{gr}}^4 - T_{\mathrm{al}}^4\right) \end{aligned} \quad (29)$$

where T_{gr} is the ground temperature, which is taken as the external ambient temperature, $h_{\mathrm{b,free}}$ and $h_{\mathrm{b,forced}}$ are heat transfer coefficients for free and forced convection, respectively, with reference to the bottom surface and are evaluated with [37].

$$h_{\mathrm{b,free}} \begin{cases} \dfrac{\left[\left(0.825 + \dfrac{0.387(\mathrm{Ra}\cdot\sin\beta)^{\frac{1}{6}}}{\left(1+(0.492/\mathrm{Pr})^{\frac{9}{16}}\right)^{\frac{8}{27}}}\right)^2\right]k_{\mathrm{a}}}{L_{\mathrm{c}}} & \text{if } \beta \ge 30° \\ \dfrac{\left[0.56(\mathrm{Ra}\cdot\sin\beta)^{\frac{1}{4}}\right]k_{\mathrm{a}}}{L_{\mathrm{c}}} & \text{if } 2° < \beta < 30° \text{ and } 10^5 < \mathrm{Ra}\cdot\sin\beta < 10^{11} \\ \dfrac{\left[0.58(\mathrm{Ra})^{\frac{1}{5}}\right]k_{\mathrm{a}}}{L_{\mathrm{c}}} & \text{if } 0 \le \beta \le 2 \text{ and } 10^6 < \mathrm{Ra} < 10^{11} \end{cases} \quad (30)$$

where the Rayleigh number Rais defined as

$$Ra = \frac{g.\left(1/T_{\mathrm{f}}\right).\left(T_{\mathrm{pv}} - T_{\mathrm{a}}\right).L_{\mathrm{c}}^3}{\nu.\alpha} \quad (31)$$

$$h_{\mathrm{b,forced}}\begin{cases}\dfrac{\left(0.664\mathrm{Re}^{\frac{1}{2}}\,\mathrm{Pr}^{\frac{1}{3}}\right)k_{\mathrm{a}}}{L_{\mathrm{c}}} & \text{for laminar flow}\\ \dfrac{\left(0.037\mathrm{Re}^{0.8}\,\mathrm{Pr}^{\frac{1}{3}}\right)k_{\mathrm{a}}}{L_{\mathrm{c}}} & \text{for turbulent flow}\\ \dfrac{\left(\left(0.037\mathrm{Re}^{0.8}-\left(0.037\mathrm{Re}_{\mathrm{c}}^{0.8}-0.664\mathrm{Re}_{\mathrm{c}}^{0.5}\right)\right)\mathrm{Pr}^{\frac{1}{3}}\right)k_{\mathrm{a}}}{L_{\mathrm{c}}} & \text{for mixed flows}\end{cases} \tag{32}$$

where Re is the Reynolds number, $\mathrm{Re}=\rho v_{\mathrm{w}}L_{\mathrm{c}}/\mu$, $v_{\mathrm{w}}=1\ \mathrm{m\ s^{-1}}$, the velocity of air. k_{a} stands for thermal conductivity of air, Pr represents the Prandtl number.

The view factors can be found from the equations,

$$F_{\text{b-sky}}=\frac{1+\cos(\pi-\beta)}{2}\quad F_{\text{b-gr}}=\frac{1-\cos(\pi-\beta)}{2} \tag{33}$$

Initial conditions are

$$u=0,\ v=0 \text{ at } t=0 \text{ and } T=27\,^{\circ}\mathrm{C} \text{ at } t=0 \tag{34}$$

Mesh generation, grid independence study and validation of the model

The model is numerically simulated using COMSOL *Multiphysics* 6.0 software based on the finite element method. The geometries of the finned PV-PCM systems are established, and relevant geometrical and thermal properties are suitably specified. The PV module within the system comprises a total of five distinct layers. The attachment of the PCM to the module's underside is facilitated by two aluminium plates. The thermal interface between the photovoltaic panel and the aluminium plate is assumed to be perfect. Table 1 outlines the specifications regarding the thickness and properties of both the module and aluminium plate layers. Table 2 outlines the specifications of properties of the PCM RT35HC.

Mesh generation and grid independent study

Physics-induced mesh is generated to carry out the simulation of the system, as shown in Fig. 2. The final mesh is selected after conducting a mesh dependence study. For this, the variation average PV temperature of three-finned PV-PCM system configurations at the inclination of 30° with time is simulated for multiple mesh element sizes, namely *coarse, normal, fine* and *extra-fine* mesh settings. The findings suggest that the *fine* and *extra-fine* type meshes give almost identical results, and the *fine* option is selected for the analysis. The final mesh comprises 86,522 domain elements and 9856 boundary elements, as shown in Fig. 3.

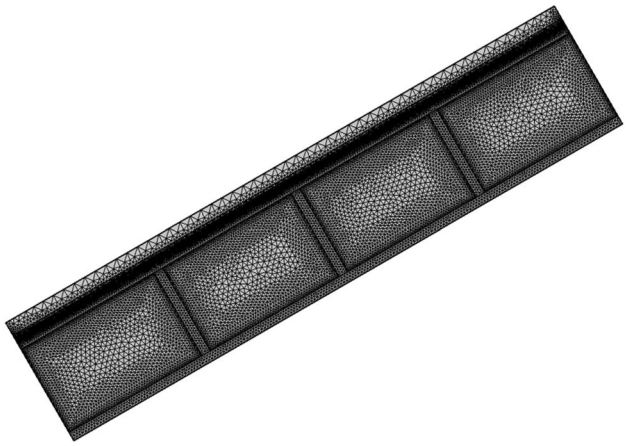


Fig. 2 Mesh generated for the simulation of the field

In COMSOL *Multiphysics*, the simulation is carried out by invoking three modules, *heat transfer in solids (*for solid PV layers and aluminium), *heat transfer in liquids* (for PCM), and laminar fluid flow (for melted PCM). Through the Boussinesq approximation, volumetric force is added to enable buoyancy-driven flow in melted PCM. The time-dependent numerical study is chosen. Backward Euler is used as the time-stepping approach for the

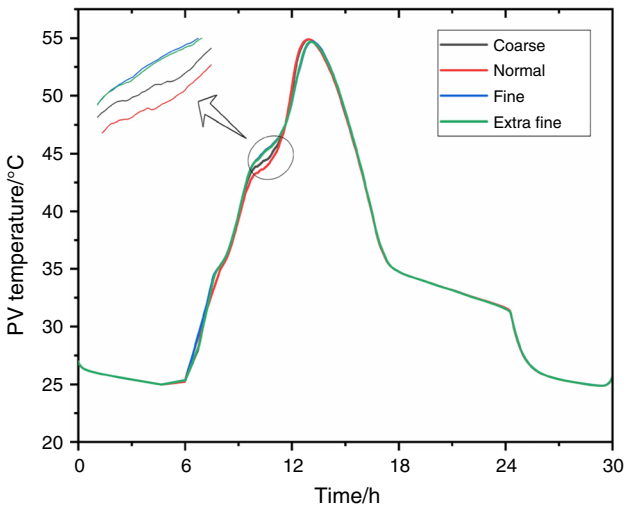


Fig. 3 Grid independence study

Table 2 Properties of RT35HC [18]

Properties	Value	
	Solid	Liquid
ρ/kg m^{-3}	880	770
c_{p}/J kg^{-1} K^{-1}	2000	2000
T_{m}/°C	35	
k/W m^{-1} K^{-1}	0.2	0.2
L_{h}/J kg^{-1}	240,000	
ν/m^2 s^{-1}		5×10^{-6}
α_{pcm}/K^{-1}	0.00091	0.00091

simulation (Backward Differentiation). The initial time step is 0.0001 s, and the maximum time step is 2 s. The absolute tolerance is specified as 0.0001 s. Performance parameters of the PV module, such as average temperature, electrical efficiency, electric power output, and performance of the PCM, such as energy storage, temperature, phase fractions, convection patterns, and average velocity, are simulated for different PV-PCM system configurations of the system. The analysis of the finned PV-PCM system is carried out for different geometrical and operating conditions in the present work.

Validation of the model

The experimental average temperature results of the finned PV-PCM system reported by Huang et al. [38] are used to validate the mathematical model considered for the simulation of the system in the present work. Initial and boundary conditions were considered for the numerical model for validation purposes to reflect the original experimental conditions. The convection current is also incorporated into the system. The interior dimensions of the systems are 40 mm wide and 132 mm high. Heat flux of 750 W m^{-2} and outdoor temperature of 23 °C are applied as conditions under which the system is in operation. The trend of the mean PV temperature with time is simulated and compared with the experimental result, as shown in Fig. 4. A notable agreement between simulation predictions and experimental results is observed, thus indicating the reliability of the proposed model in simulating the system with sufficient accuracy.

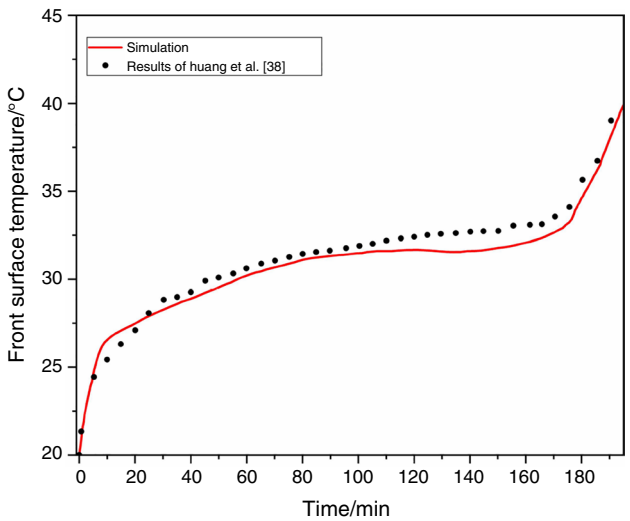


Fig. 4 Validation of the model with the experimental results

System simulation

Different configurations of the system are obtained by considering the number of fins in the system as 0 (no fin), 1, 2, 3, and 4, and for a specific fin case, the inclination of the system under consideration is 0°, 15°, 30°, and 45°. The simulation is carried out for all the configurations of the system with the same input profiles of solar radiation and ambient temperature for 2 days in the City of Kozhikode, Kerala, India, on a particular day [39] as shown in Fig. 5, using a 12-core computer with 32.00 GB of RAM. The input parameters required to carry out the simulation of all the configurations of the system are shown in Table 3.

Results and discussion

For unfinned system

The time history of the average PV temperature of all simple PV-PCM configurations (with no fin) is simulated and depicted in Fig. 6. At first; the PV temperature slightly

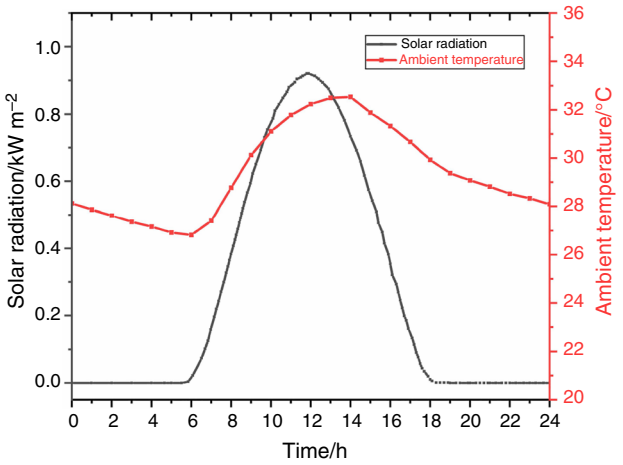


Fig. 5 Variation of solar irradiation and ambient temperature with time

Table 3 Inputs for simulation

Parameter	Value
Initial temperature/°C	27
height of fin/mm	20
height of PCM/mm	20
$(\tau\alpha)$	0.9
Wind azimuth angle, γ/°	30
ε_{al}	0.85
ε_{gl}	0.92
Wind speed, v_w/m s^{-1}	1
ΔT/°C	2

decreases with time (from 0.00 to 6.00 h) due to the absence of incoming solar radiation to the system and the heat transfer to the surroundings from the system by the modes of convection and radiation. From 6.00 h onwards, the PV temperature increases with time with incident solar radiation and attains a maximum value at around 13.30 h. A minor fraction of incident solar radiation is converted to electrical energy by the PV module, and the major fraction is wasted as heat, increasing the module's temperature. In the current PV-PCM setup, PV temperature is regulated by virtue of heat transfer from the module to the PCM. The rate of conversion of incident radiation to thermal energy dominates the heat transfer rate from the module to the PCM and surroundings, resulting in a rapid increase in module temperature from 6.00 to 12.00 h for the horizontal configuration of the simple PV-PCM system. This is due to (1) low heat transfer by conduction mechanism from the module to PCM by virtue of poor thermal conductivity of paraffin PCM apart from the heat transfer by the module to surroundings by radiation and convection and (2) increasing trend of incoming radiation with time until 12.00 h.

The increasing trend of the mean temperature of the PV continues till 13.30 h despite the decreasing trend of incident radiation with time from 12.00 to 13.30 h. This is due to the persistent dominance of the fraction of thermal energy converted from incident radiation over the heat transfer to PCM and surroundings in this duration under the prevailing thermal conditions in the system. The temperature and thermal energy of the PCM increase with time. However, the increase in temperature mainly occurs when it is in a single phase. Energy is stored in the PCM in sensible form and the combination of sensible and latent forms when it is in a single phase and during phase transition, respectively.

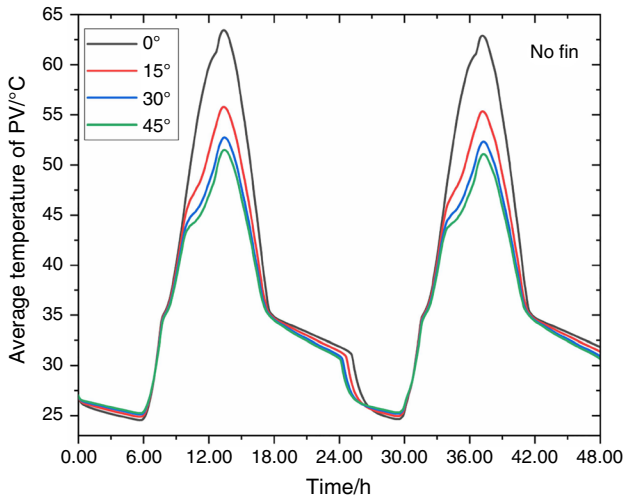


Fig. 6 Variation of the average temperature of PV with the time of PV-PCM system for various inclinations

The magnitude of energy stored in the form of latent energy is very high compared to sensible energy. The PCM starts melting around 8.30 h. The mean temperature of PV began to decrease around 13.30 h with time due to the decrease in the rate of thermal energy available to it with time by virtue of the decreasing trend of the incident solar radiation. At 16.00 h, PCM begins to solidify, releasing latent heat and changing to the solid phase. The cycle then continues for the next 24 h.

When the inclination of the system is 0° (horizontal position), due to the existing density variation in the fluid phase of the PCM, the melted fluid phase of the PCM is stable, and no natural convection current is set up in it. As the inclination angle increases, the viscous force within the fluid can no longer sustain the buoyancy forces, and convective motion is set up in the PCM, which gives rise to circulation patterns in the fluid phase of the PCM. The trends of the average temperature variation of the PV with time for inclined configurations are almost the same as for horizontal configurations initially, as conduction is the main heat transfer mechanism in both configurations. The slope of the mean temperature trend decreases for the inclined configuration when the phase change process commences due to a higher heat transfer rate from the PV module to the PCM, as natural convection heat transfer also comes into play within the PCM and from the system to its surroundings. The higher the system's inclination, the greater the heat transfer from the module to PCM and surroundings, and the lower the slope of the mean temperature trend. A lower slope of the mean temperature trends can be observed for inclined configurations throughout the phase change process than for horizontal configurations.

The temperature contours of the simple PV-PCM system at different instants and various inclinations are shown in Fig. 7. Arrows represent velocity vector plots that depict the advancement of the melted PCM. For the horizontal PV-PCM configuration, the solid–liquid interface inside the PCM exists in a plane parallel to the plane in which the system exists and moves approximately linearly from top to bottom with time. As the heat transfer inside the PCM is primarily due to conduction, the rate of energy stored in the PCM is lower, leading to a high PV temperature. For an inclined PC-PCM configuration, buoyancy force plays a significant role in liquid PCM and takes an active role in heat transfer to the PCM, and influences the solid–liquid interface movement. The orientation of the interface is also similar to the plane of the system's existence for inclined configurations. Increasing the inclination angle leads to greater buoyancy-induced flow and a greater heat transfer rate to the PCM.

For an inclined PV-PCM, low-density melted PCM close to PV moves upward, and cooler PCM with high density fills the space, thereby forming a natural convection current.

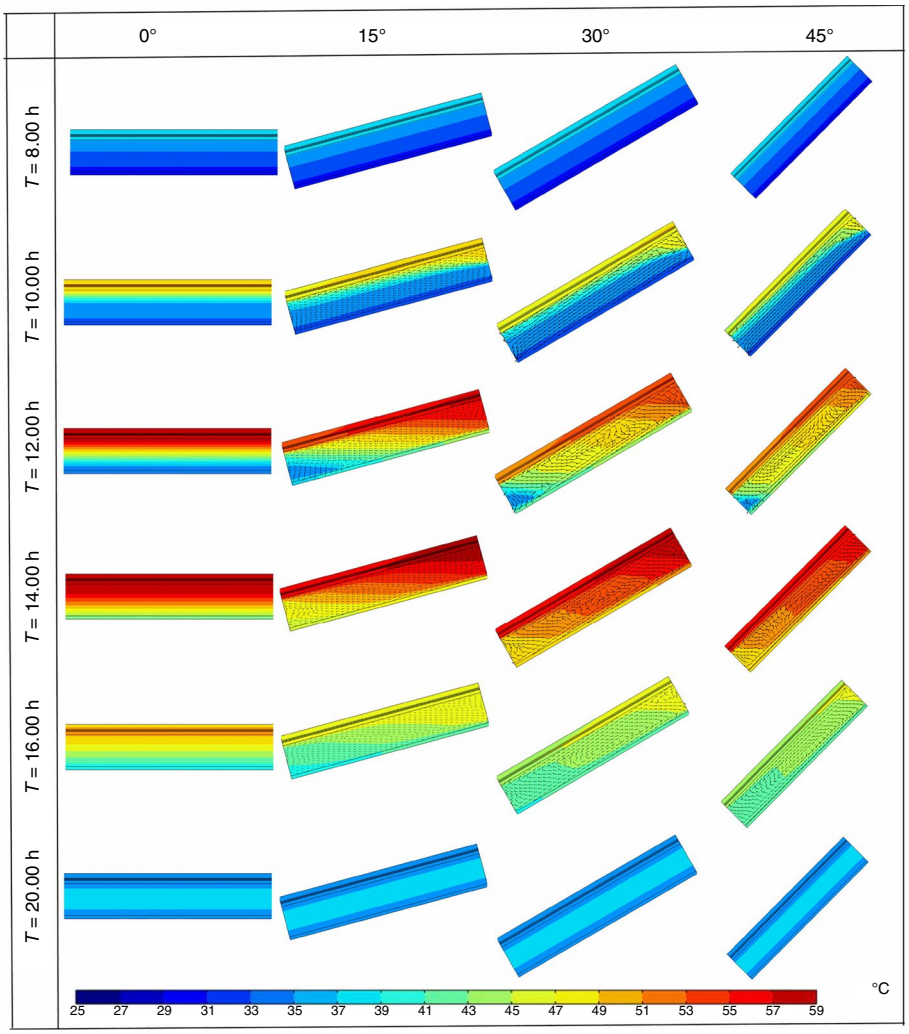


Fig. 7 Temperature contours and velocity vectors for a simple PV-PCM system (with no fin) at different instants for different inclinations

The direction and magnitude of the natural convection currents can be seen with the direction and magnitude of arrows since the fluid flow is represented by typical velocity vector plots. Due to the non-existence of convection currents for PCM in the horizontal configuration, the heat transfer rate to the melted PCM is less compared to inclined PCM configurations. For an inclined PV-PCM system, the existence of buoyant force leads to natural convection heat transfer in the fluid PCM, and as a result, the melting rate of PCM increases. When the inclination angle of the system increases, the strength of the convection current increases, leading to higher circulation in the fluid PCM, which enhances the rate of heat transfer. Hence, PV's highest and lowest mean temperatures are at 0° and 45° inclination angles. The magnitude of the mean temperature is its maximum, around 13.30 h. During solidification, the natural convection heat transfer rate decreases, and heat transfer mainly takes place due to conduction.

The instantaneous electrical power output from PV at different instants is compared for the different configurations of the un-finned PV-PCM system, as given in Fig. 8. The electric output is calculated using the equation, $P_{out}=\eta_{PV}(\tau\alpha)A_s G_{module}$. The power output from the PV module increases with time from 6.00 h. It reaches a peak and, after that, decreases with time in all the configurations of the PV-PCM system, which is similar to the trend of incident radiation with time. The output is almost the same for all the inclinations up to 8.00 h, as the effect of natural convection current is negligible till that time for all the configurations. Electrical output is found to increase with the increase in inclination at 10.00 h, 12.00 h, and 14.00 h due to the presence of a natural convection mechanism in the melted PCM in all configurations except the horizontal configuration. The increase in strength of convection currents and circulation with inclination leads to a lower PV temperature at higher inclinations at that instants. The peak output of 159.7 W is obtained for the system with $\beta = 45°$.

The variation in PV electric efficiency with time during the day (from 6.00 to 18.00 h) for different configurations of unfinned PV-PCM systems is shown in Fig. 9. It is observed

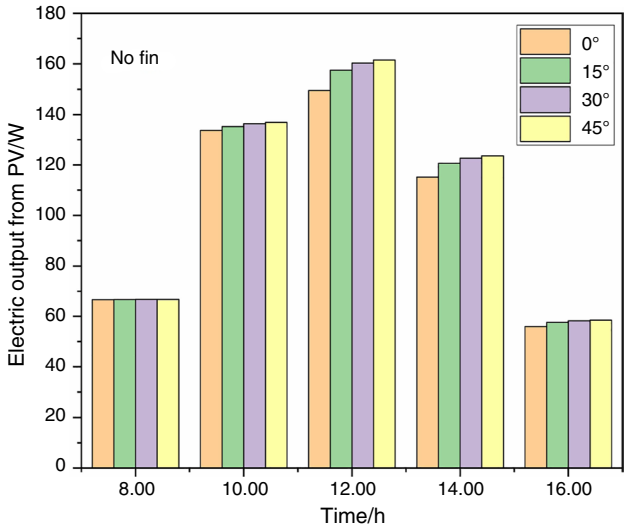


Fig. 8 Variation of instantaneous electric output from PV module for unfinned PV-PCM systems for different inclinations

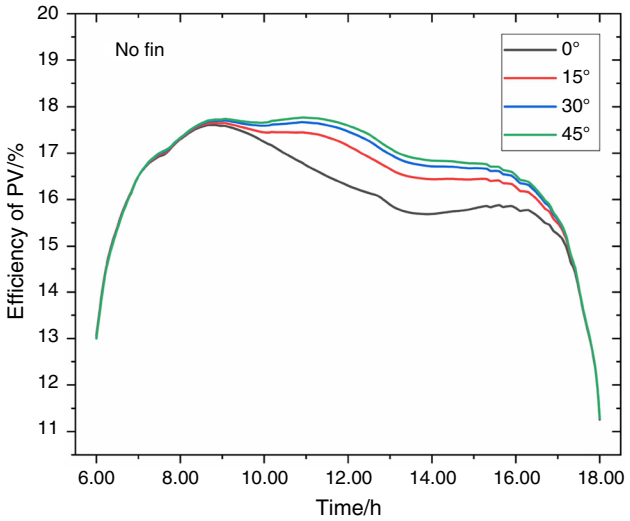


Fig. 9 Variation of PV efficiency for un-finned PV-PCM configurations at different inclinations

that efficiency is initially low and then increases with time as solar radiation has a positive effect on PV efficiency. Up to 9.00 h, efficiency is the same for inclinations, and after that, efficiency decreases with time due to the increase in PV temperature. The higher the average temperature, the lower the efficiency. So, efficiency is minimal for the inclination of 0° due to the high mean temperature compared to the configurations of other inclinations. Efficiency decreases with time for all the configurations in the evening because of the decrease in irradiation, even though the average PV temperature decreases with time in this duration.

For finned systems

The performance of finned PV-PCM configurations with the variation of inclination is simulated for four cases of the system (number of fins = one, two, three and four, placed equidistantly along the length of the PCM container with the height equal to the depth of PCM). The PV temperature is anticipated to decrease with the finned PCM configurations compared to the unfinned system. This is attributed to the larger surface area provided by the fins, which enhances the heat transfer from the PV module to the PCM. The inclusion of fins significantly enhances the thermal conductivity of the subsystem, as well as promoting flow mixing within the liquid phase of the PCM. The average temperature variation of the PV module for finned PV-PCM configurations is shown in Fig. 10. The average temperature of PV reduces considerably with the presence of fin(s). For all finned cases, it can be observed that there is a decrease in the peak average PV temperature as the inclination angle increases.

The temperature contours and velocity vectors of different finned PV-PCM configurations are shown in Fig. 11a–d. Here, the PCM begins to melt early compared to unfinned PCM configurations (around 7.30 h). As the number of fins increases, the PV temperature reduces further as more surface area is available for heat transfer, increasing the conduction heat transfer rate to the PCM. However, with the increased number of fins, a penalty exists in the form of a weakened convection current in the melted PCM due to increased resistance to the flow and decreased buoyancy force. This decreases the convection heat transfer rate with an increase in the number of fins. The highest reduction in temperature due to the inclusion of fin(s) is observed for horizontal PV-PCM configurations; as here, only conduction is the main heat transfer mechanism within the PCM, and the physical situation of increase in resistance with an increase in surface area will not arise. The influence of the inclination angle on the temperature of the PV is significant up to the number of fins, $n = 2$. For a higher number of fins, the convection current is weakened due to increased flow resistance and decreased space available for flow circulation imposed by the fins, thereby decreasing convection heat transfer within melted PCM, although there is an increase in conduction heat transfer to it.

Trends of electrical output from PV at different instants for finned configurations are shown in Fig. 12. The electrical output for finned PV-PCM is higher compared to unfinned PV-PCM configurations. The influence of inclination on electrical output can also be observed at 10.00 h, 12.00 h, and $t = 14.00$ h. Figure 13 shows the PV efficiency variation during the daytime for all finned PV-PCM configurations with inclination. The effect of the inclination angle can be felt from 9.00 h onwards, where natural convection

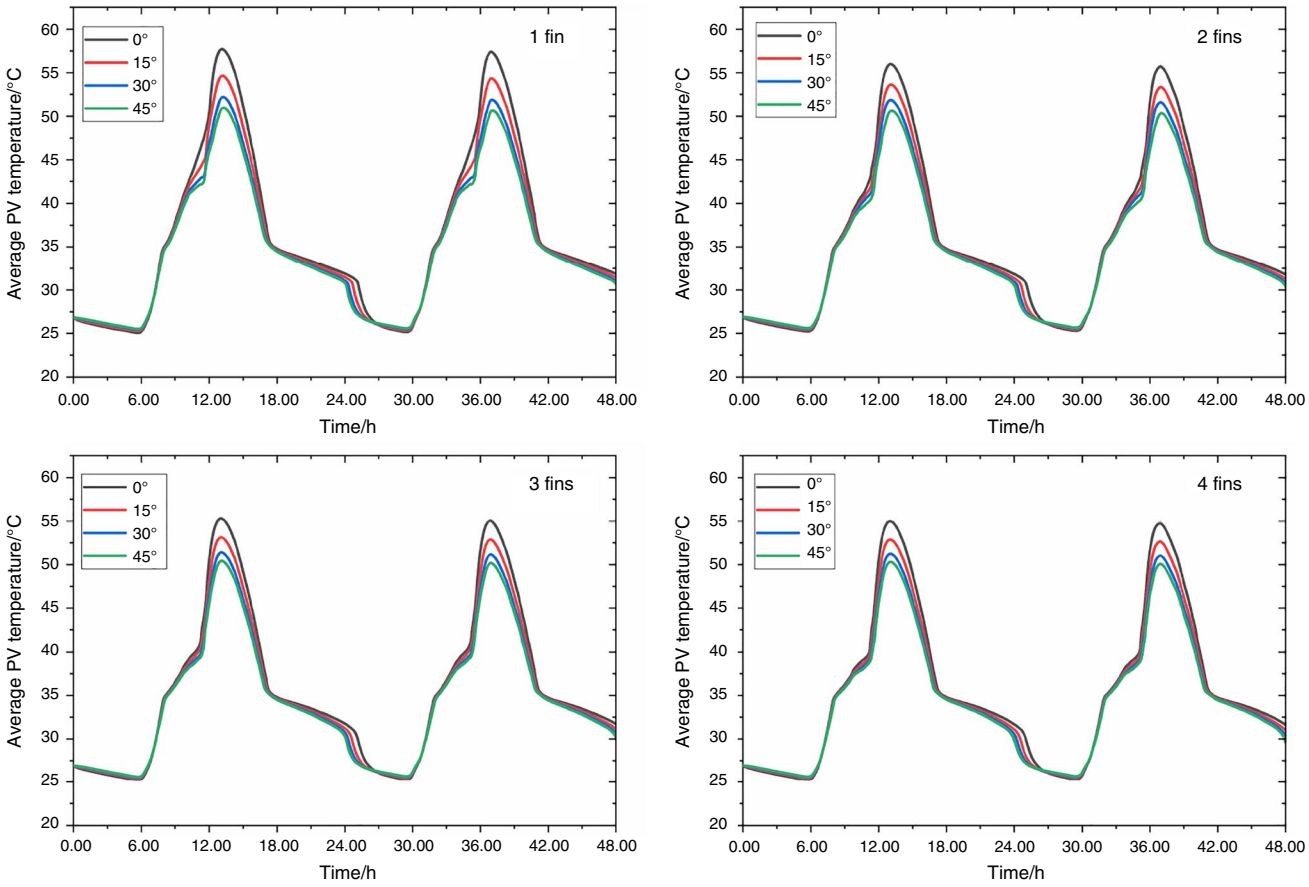


Fig. 10 Variation of the average temperature of PV with time of finned PV-PCM configurations for various inclinations

becomes active in melted PCM. Maximum variation is found at around 13.00 h, where the strength of the convection current is maximum.

The peak average temperature of PV for all the configurations considered for simulation is given in Fig. 14. The simple PV-PCM system with an inclination of 0° has the highest peak temperature, and it decreases with an increase in inclination due to enhanced natural convection heat transfer to the PCM. The natural convection heat transfer in liquid PCM plays an essential factor in the thermal regulation of the PV module for the un-finned PV-PCM configurations. The increase in inclination decreases the peak temperature of PV for all the finned PV-PCM configurations. However, the percentage reduction in peak temperature is less than in the unfinned PV-PCM configurations. With fins, conduction heat transfer increases due to the increased heat transfer area, but there is an increase in resistance to the flow of liquid PCM, adversely impacting natural convection circulation. This leads to a situation in which a proportionate improvement in the thermal regulation of PV and the rate of energy stored in the PCM could not be obtained in finned PV-PCM configurations. There is no significant variation in peak temperature when the number of fins exceeds 3. The peak temperature for the horizontal un-finned PV-PCM configuration is 63.45 °C, which reduces to 54.45 °C when the inclination is 45°. The lowest peak temperature of PV is 50.23 °C for the PV-PCM configuration with three fins at $\beta = 45°$.

Figure 15 compares the maximum enhancement in PV efficiency at different numbers of fins and various inclinations of the system. The efficiency enhancement is highest for the PV-PCM system with $\beta = 0°$ for any number of fins. When the system is in the horizontal position, i.e. $\beta = 0°$, the dominant heat transfer mechanism is conduction and will be enhanced with the addition of fins. For inclined systems, convection is the dominant heat transfer mechanism, and convection current decreases with the number of fins as the resistance to bulk flow for melted PCM increases as the flow channel's length is reduced. For $\beta = 0°$, the efficiency enhancement is 3.4% for one fin and 5.56% for four fin systems when compared to corresponding no-fin PV-PCM systems; meanwhile, for 45°, enhancement values are 1.39% and 2.51% for one fin and four fin systems, respectively.

As explained earlier, the buoyancy force in the liquid PCM gets weakened with the increase in the number of fins, which implies less strength in the natural convection

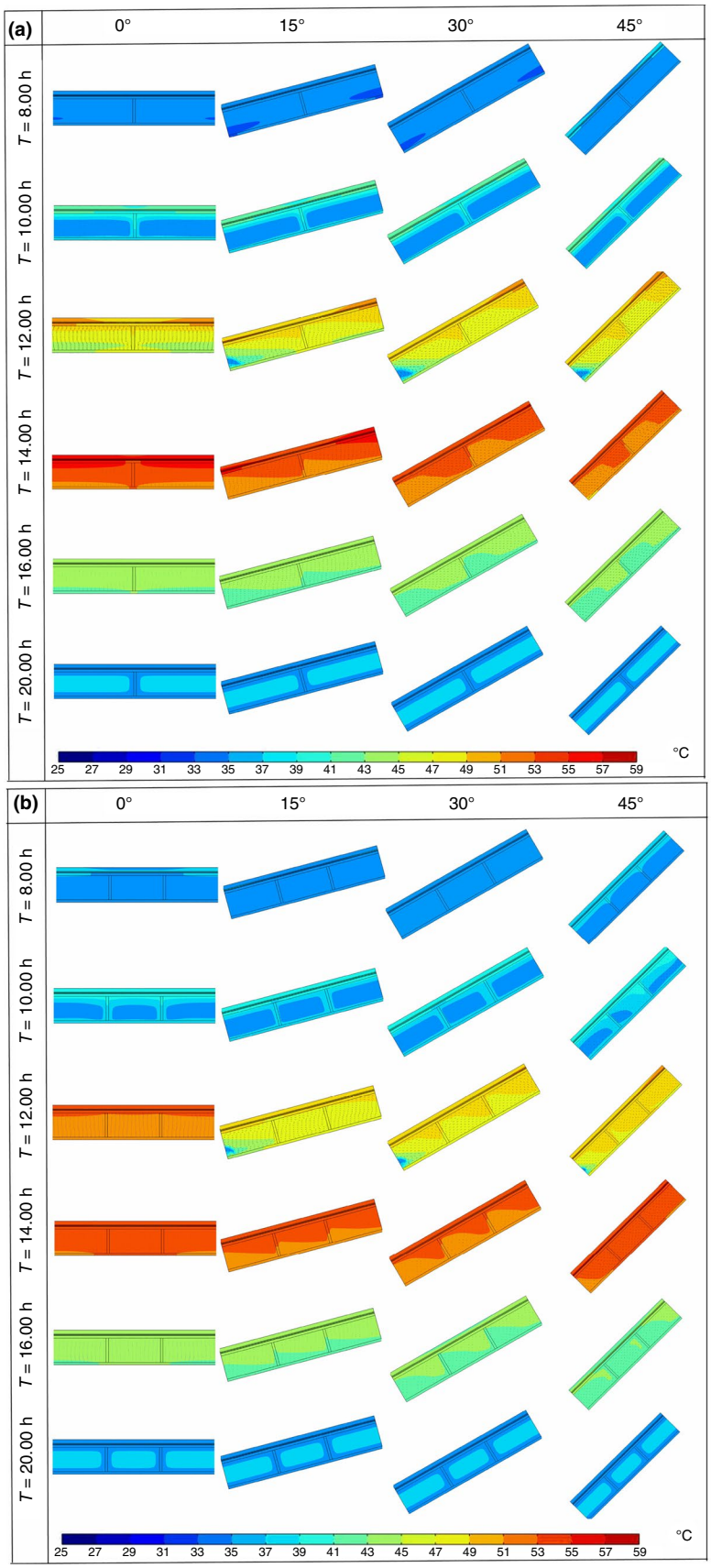
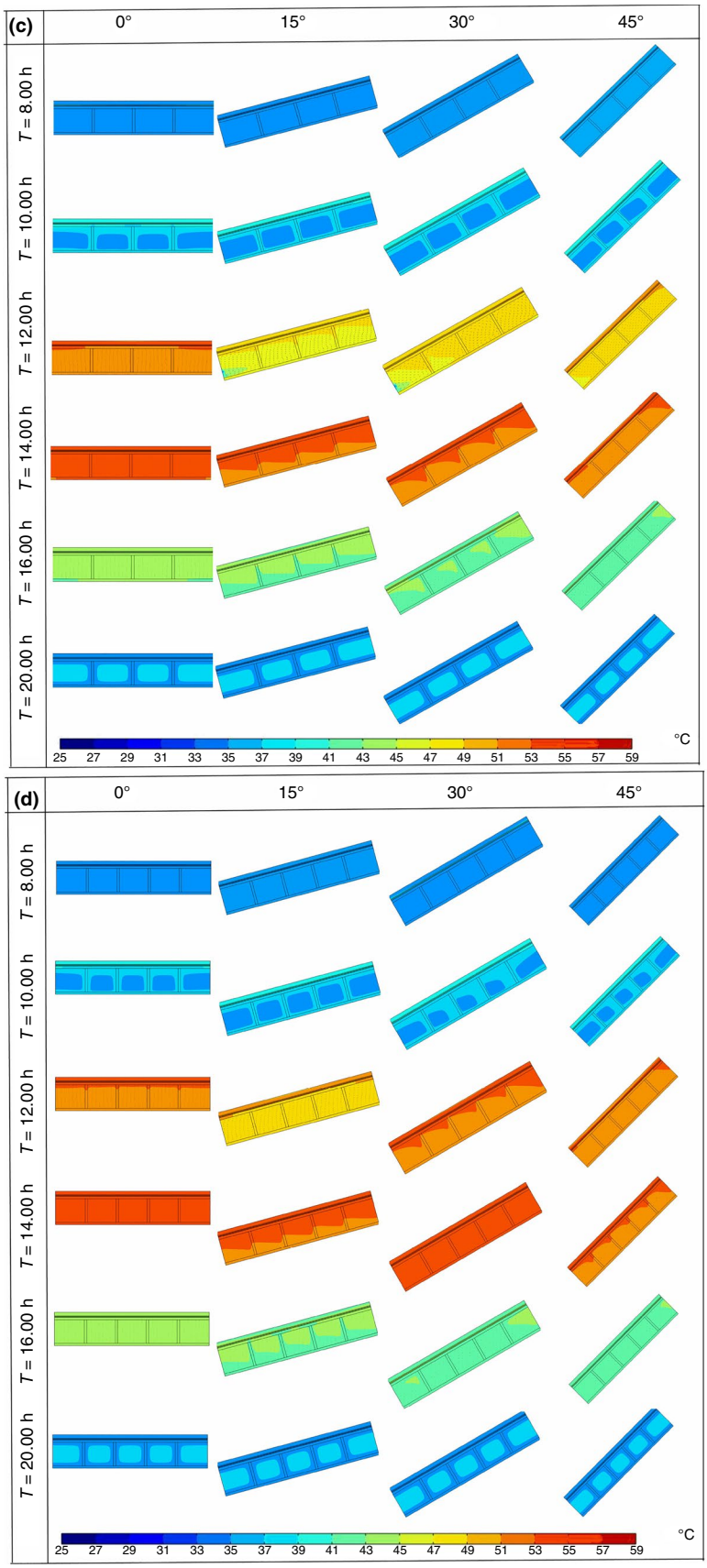


Fig. 11 Temperature contours and velocity vectors for finned PV-PCM systems with number of fins **a** $n=1$, **b** $n=2$, **c** $n=3$, and **d** $n=4$ at different instants for different inclination

Fig. 11 (continued)



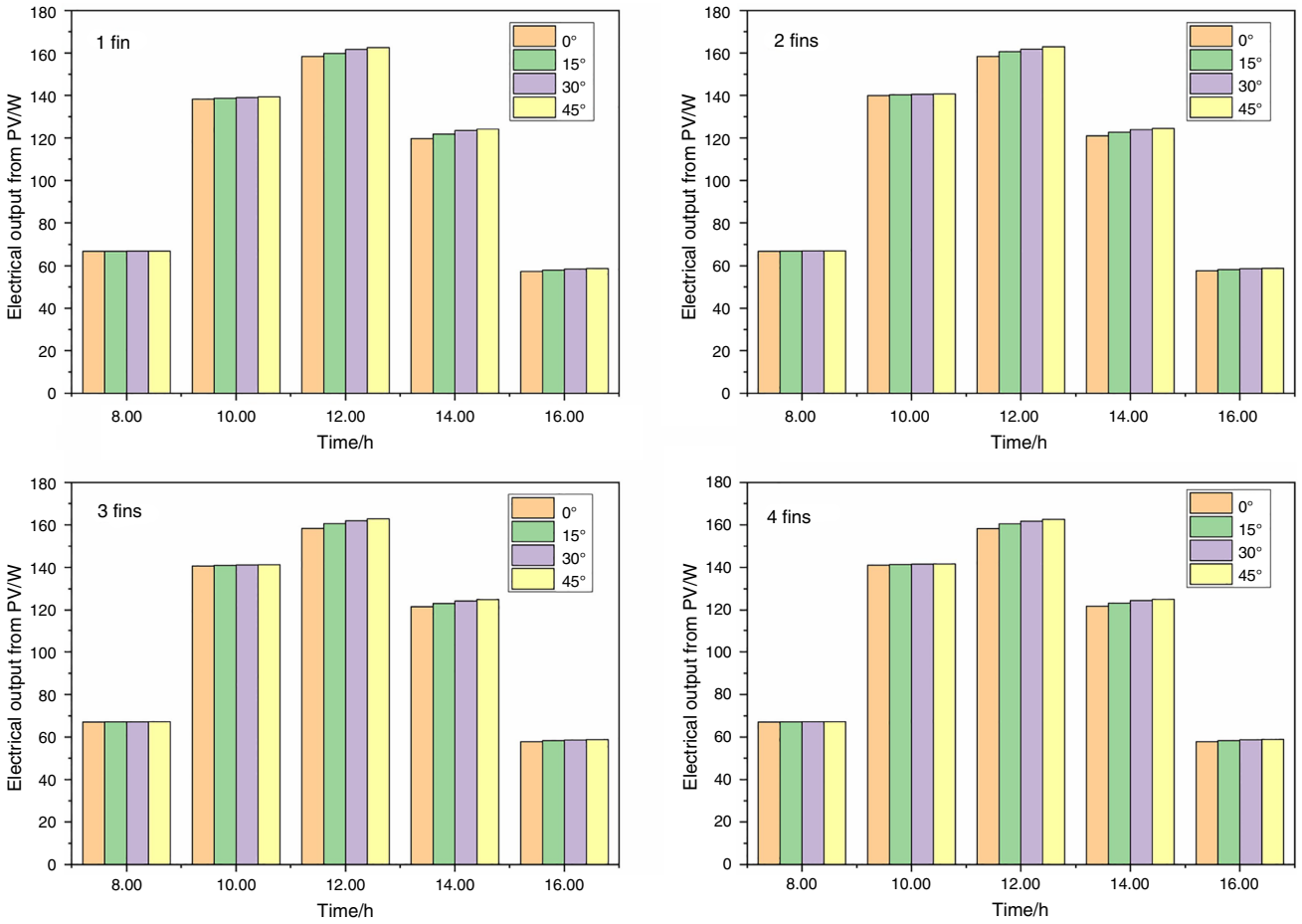


Fig. 12 Variation of instantaneous electric output from PV module for finned PV-PCM system with the inclination

current. It is clear from Fig. 16 that the peak mean velocity increases with an increase in the inclination for a particular number of fins and decreases with the increase in the number of fins for a particular inclination, except for the horizontal configuration without fins. At 0° inclination, unfinned PV-PCM has almost zero mean velocity ($\approx 5.25 \times 10^{-5}$ mm s^{-1}) due to the existence of a very weak buoyancy force. For inclined unfinned configurations, natural convection currents exist in the liquid PCM during the phase change process, and the strength of the natural convection current proportionately increases with an increase in inclination, which increases the peak mean velocity with inclination in these configurations. The relative increase in peak mean velocity with the increase in inclination angle is very high for unfinned configurations since these configurations are accompanied by minimum resistance to flow. The maximum velocity, 0.794 mm s^{-1}, is found for the unfinned PV-PCM case with an inclination of 45°.

In contrast to horizontal unfinned configurations, nonzero peak mean velocity exists for finned horizontal configurations. This is due to the change in PCM domains with the introduction of the fins in the system for these configurations. The existence of unbalanced force in the domains of horizontal finned configurations leads to the initiation of small natural convection currents, which cause a natural convection heat transfer mechanism during the phase transition. The higher degree of natural convection heat transfer during the phase change process of the PCM exists in all inclined finned PCM configurations. Resistance to flow increases with the increase in the number of fins at a particular inclination due to the reduction in the size of the fluid flow domain, which weakens the natural convection current and reduces the peak mean velocity with the increase in the number of fins.

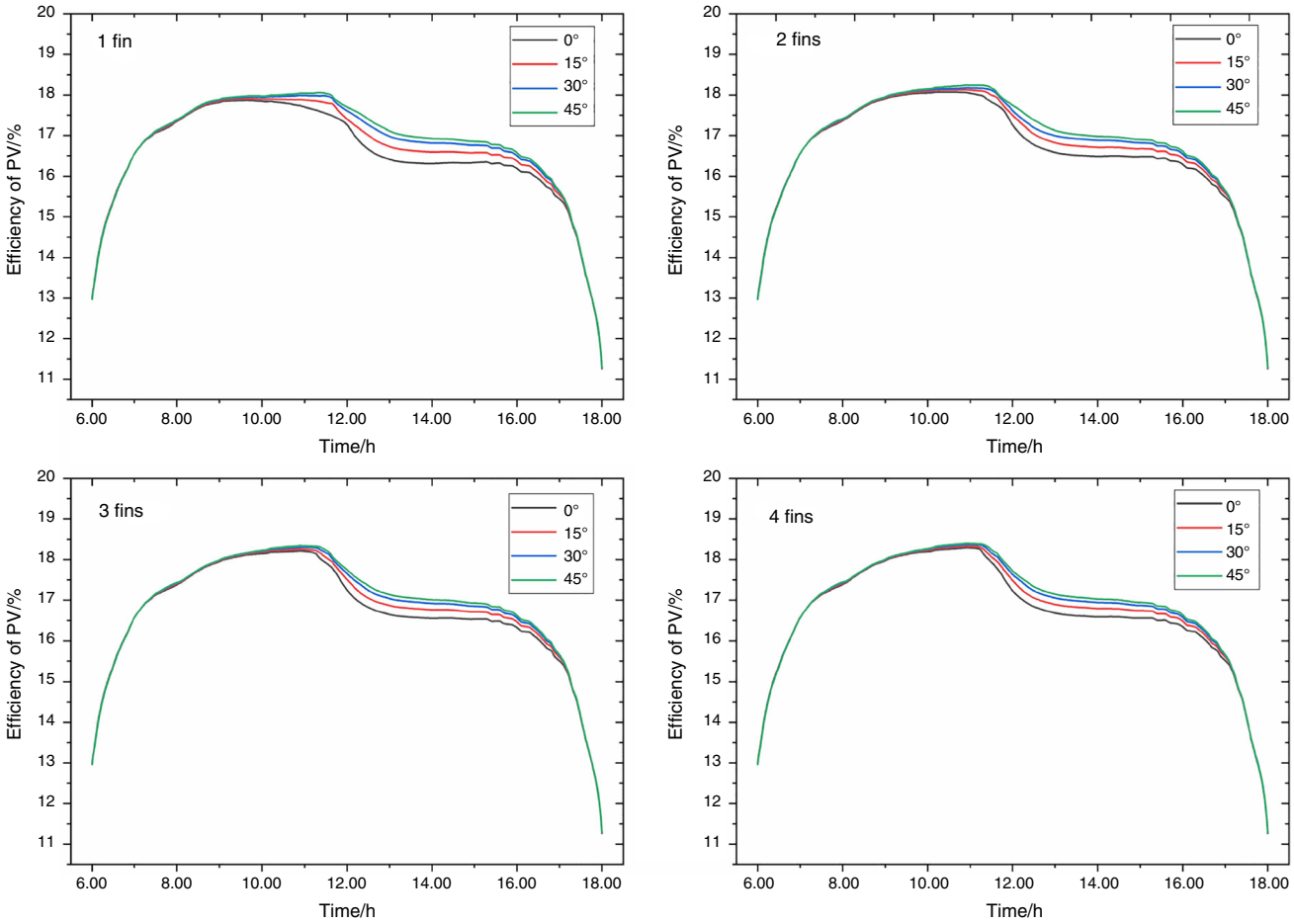


Fig. 13 Variation of electrical efficiency of PV module during day time for finned PV-PCM configurations at different inclinations

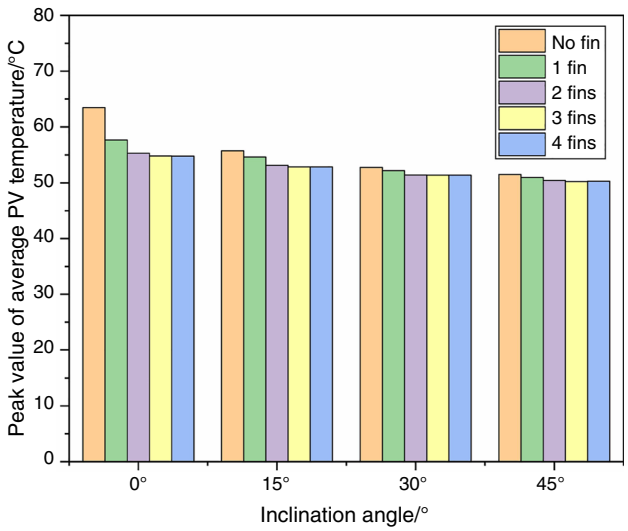


Fig. 14 The effect of the number of fins and inclination on peak mean temperature of PV for all PV-PCM systems

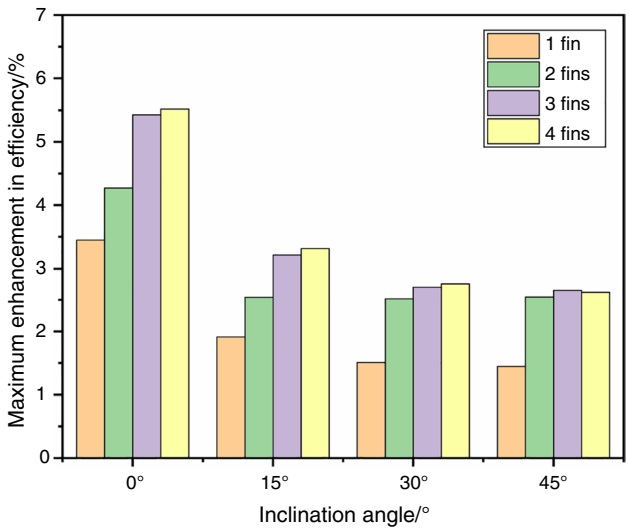


Fig. 15 Comparison of maximum enhancement in PV efficiency for all finned PV-PCM systems at different inclinations

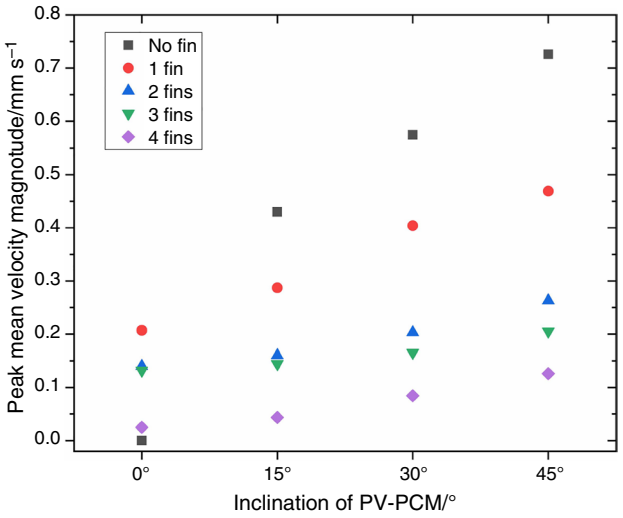


Fig. 16 Effect of the number of fins and inclination on the peak mean velocity of liquid PCM for all configurations

Conclusions

The combined effect of the number of fins and inclination on the performance of the finned PV-PCM system was studied. The findings of the current investigation are:

- For the unfinned and finned PV-PCM systems with an inclination of 0°, heat transfer inside the PCM occurs due to the conduction mechanism. The increase in the number of fins further increases the heat transfer inside PCM and reduces PV temperature.
- For inclined unfinned PV-PCM systems, convection is the major heat transfer mechanism in the liquid PCM because of the existing thermal and velocity fields. The degree of convection increases with an increase in inclination.
- The addition of fins to inclined finned PV-PCM systems improves conduction heat transmission within the PCM. Smaller domains of the liquid PCM existing with the inclusion of fins in the system increase the resistance to bulk motion. This contributes to a decrease in the convection heat transfer rate.
- The inclusion of fins is relatively most effective for the PV-PCM system at $\beta = 0°$ compared to the configurations at other inclinations. The highest percentage improvement in PV efficiency is 5.56%, which is found for the PV-PCM system with four fins and $\beta = 0°$.
- Considering the trends of average PV temperature and PV efficiency, the three-finned PV-PCM system with an inclination of 45° is found to have the lowest temperature and the highest electric efficiency. Therefore, this configuration is the optimum configuration.
- The highest and lowest peak average PV temperatures of 63.45 °C and 50.23 °C are obtained for the unfinned PV-PCM system at $\beta = 0°$ and the three-finned PV-PCM system at $\beta = 45°$, respectively.
- The maximum instantaneous PV electrical power output of 165.87 W is obtained for the three-finned PV-PCM system with an inclination $\beta = 45°$, compared to 159.5 W obtained for unfinned PV-PCM for the same inclination.
- The mean velocity of liquid PCM rises with an increase in the inclination for a particular number of fins and decreases with the increase in the number of fins for a particular inclination, except for horizontal configuration without fin.
- The peak mean velocity of the liquid PCM is highest for unfinned PV-PCM systems with an inclination of 45°. The magnitude of the velocity of melted PCM indicates the strength of the convection current. For $\beta = 45°$, the peak mean velocity magnitude drops from 0.794 mm s^{-1} for no fin case to 0.162 mm s^{-1} for four fin case.
- The increase in fin number in the PV-PCM system beyond three results in scanty benefit in terms of thermo-electric performance. In this study, three-finned PV-PCM system with an orientation of $\beta = 45°$ is found to be the best one.

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Author contributions Unnikrishnan K S has contributed to the main conceptual idea, study design and full article writing. Guduru Prudhvinath, Atti Sai Praneeth, Gauvrav Singh and Jayant Prakash helped in conducting simulations and article writing. Rohinikumar B has been involved in the technical discussions, supervising the whole work, checking the review connectivity and improving the write-up. All authors read and approved the final manuscript.

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Data availability The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

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